

into the directory */log*, and log the packets relative to the *192.168.1.0* class C network. All incoming data packets will be recorded in the sub-directories of the main directory log with the names of the directory based on the address of the remote host. If you are working on a high speed network or want to log the packets in a compressed form for later analysis, consider logging in 'binary mode'.

In the binary mode, the logging of the packets is done in the 'tcpdump format' to a single binary file in the logging directory:

```
./snort -l ./log -b
```

We should note that we have not specified a home network in the command above because the binary mode logs everything to a single file, in which there is no need to specify how to format the output directory structure.

Network intrusion detection mode

To run the tool in network intrusion detection (NIDS) mode, execute the following command:

```
./snort -dev -l ./log -h 192.168.1.0/24 -c snort.conf
```

snort.conf is the name of the file that has different rules. The rules set is applied from the *snort.conf* file to each packet to finally decide whether or not to take a particular action based upon the rule type in the file. If we do not specify the output directory for the program, it will be */var/log/snort* by default.

Reading PCAP using Snort and alert files

PCAP files can be analysed very easily using Snort. The PCAP file is passed with a Snort command. Once the command is executed, Snort generates the alert file from that specific PCAP file.

To read a single PCAP file, use the following commands:

```
$ snort -r mynetwork.pcap
$ snort --pcap-single= mynetwork.pcap
```

To read multiple PCAPS from a file, use the following:

```
$ cat mypcaps.txt
pcap1.pcap
pcap2.pcap
$ snort --pcap-file=mypcaps.txt
```

This command will read *pcap1.pcap*, *pcap2.pcap* and all files under */home/mynetwork/pcaps*. Snort will not attempt to check whether the files under that directory are really PCAP files or not.

To read PCAPS from a command line list, use the following command:

```
$ snort --pcap-list="MyNetwork1.pcap MyNetwork2.pcap MyNetwork3.pcap"
```

This will read *MyNetwork1.pcap*, *MyNetwork2.pcap* and *MyNetwork3.pcap*.

To read PCAPS under a directory, type:

```
$ snort --pcap-dir="/home/MyNetwork/pcaps"
```

This will include all of the files under */home/MyNetwork/pcaps*.

You can also use filters to read a PCAP file:

```
$ cat MyNetwork.txt
MyNetwork1.pcap
MyNetwork2.pcap
```

Here, the current directory is */home/MyNetwork/pcaps*

```
$ snort --pcap-filter="*.pcap" --pcap-file=MyNetwork.txt
$ snort --pcap-filter="*.pcap" --pcap-dir=/home/MyNetwork/pcaps
```

The above will only include files that match the shell pattern **.pcap*; in other words, any file ending in **.pcap*.

```
$ snort --pcap-filter="*.pcap --pcap-file=MyNetwork.txt \
> --pcap-filter="*.cap" --pcap-dir=/home/MyNetwork/pcaps
```

In the above, the first filter **.pcap* will only be applied to PCAP files in the file *MyNetwork.txt* (and any directories that are recursed in that file). The addition of the second filter **.cap* will cause the first filter to be forgotten and then be applied to the directory */home/MyNetwork/pcaps*; so only files ending in **.cap* will be included from that directory.

```
$ snort --pcap-filter="*.pcap --pcap-file=MyNetwork.txt \
> --pcap-no-filter --pcap-dir=/home/MyNetwork/pcaps
```

In this example, the first filter will be applied to *MyNetwork.txt*. Subsequently, no filter will be applied to the files found under */home/MyNetwork/pcaps*; so all the files found under it will be included.

```
$ snort --pcap-filter="*.pcap --pcap-file=MyNetwork.txt \
> --pcap-no-filter --pcap-dir=/home/MyNetwork/pcaps \
> --pcap-filter="*.cap" --pcap-dir=/home/
```